

RELAY-X

THE HANDOVER FIREWALL

4-HOUR HACKATHON BUILD PLAN

Core pitch: “Most AI summarizes what was said. RELAY-X decides what is safe to pass forward.”

Build one unforgettable capability: a voice agent that listens to a simulated clinical handover, tracks the current truth state, detects corrections/conflicts, stops stale speech, verifies the authoritative value, and relays only the verified state to the receiving clinician.

Hackathon safety: simulated/demo data only. No diagnosis, prescribing, or autonomous treatment decisions.

1. WHAT YOU ARE BUILDING TODAY

Component	Today's MVP
Voice	Microphone → streaming STT → LLM/state logic → Rime voice output
Truth State	HEARD / CORRECTED / VERIFIED / SUPERSEDED / CONFLICTED
Safety Rule	Unverified or superseded facts cannot enter the final handover
Barge-in	Human correction stops current speech and invalidates stale output
Conflict Resolver	Two different values → ask which value is authoritative
Dashboard	Live transcript + Truth State + blocked/verified facts + final relay

2. THE KILLER FEATURE: STALE RESPONSE KILL SWITCH

The AI is speaking an old value. The clinician interrupts with a correction. RELAY-X immediately cancels the active spoken response, marks the old value SUPERSEDED, updates the canonical state, and refuses to relay the stale value.

Paramedic: “Patient received 10 mg morphine.”

System: Morphine = 10 mg [HEARD]

AI starts: “The patient received 10—”

Paramedic: “WAIT. It was 5 mg.”

System: TTS CANCELLED → 10 mg SUPERSEDED → 5 mg CORRECTED

AI: “I heard 10 mg and then 5 mg. Should I use 5 mg as the final value?”

Paramedic: “Yes.”

System: 5 mg VERIFIED → RELAYED

Judge takeaway: Humans can mis-speak. STT can mishear. AI output can become stale. The next team must still receive the latest verified state.

3. TRUTH-STATE MODEL

Status	Meaning	Can relay?
HEARD	Explicitly detected but not confirmed	NO
INFERRED	System inferred it; not explicitly verified	NO
CORRECTED	Human supplied a newer value	NO — verify first
CONFLICTED	Two values disagree	NO — resolve
VERIFIED	Human explicitly confirmed the value	YES
SUPERSEDED	Older value replaced by newer value	NO

Minimal state object

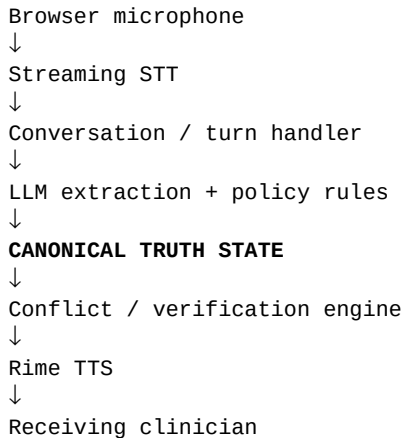
```
{
  "allergy": { "value": "none", "status": "verified" },
  "morphine": {
    "value": "5 mg", "status": "verified",
    "history": [ { "value": "10 mg", "status": "superseded" } ]
  }
}
```

Non-negotiable rule: HEARD $\neq$ VERIFIED.

4. 4-HOUR EXECUTION TIMELINE

Time	Build	Done when...
0:00–0:30	Skeleton + environment	Frontend/backend starts; config/API keys work
0:30–1:15	Voice loop	Speak → STT → LLM → Rime response works
1:15–2:00	Truth State	Facts become structured JSON with statuses
2:00–2:45	Conflict engine	Old/new values are detected and reconciled
2:45–3:20	Stale kill switch	Current speech stops; stale value is blocked
3:20–3:45	Dashboard	Judge sees transcript, state, conflict, final relay
3:45–4:00	Rehearse	90-second demo works repeatedly

5. SIMPLE ARCHITECTURE



Architecture rule: the application owns the canonical state, status transitions, relay gate, and stale-output protection. The LLM helps extract and phrase information; it does not decide which stale fact is allowed through.

6. ONLY THESE API FUNCTIONS

Function	Purpose
POST /session	Create simulated handover session
POST /utterance	Accept transcript text if streaming voice is unstable
POST /extract	Extract fields into structured state
POST /verify	Mark a field verified after explicit confirmation
POST /resolve	Resolve conflicting values
GET /state	Return current Truth State
POST /relay	Generate final handover from VERIFIED facts only

7. DASHBOARD — DO NOT OVERBUILD

LEFT — LIVE CONVERSATION

Streaming transcript with latest utterance highlighted.

CENTER — TRUTH STATE

Allergy: None — VERIFIED

Morphine: 5 mg — VERIFIED

Morphine: 10 mg — SUPERSEDED

BP: 90/60 — HEARD

RIGHT — RELAY

Only verified facts appear in the final handover.

TOP BAR: Listening | Verifying | Conflict detected | Safe to relay

The dashboard must make state transitions visually obvious. The judge should understand the product without reading your code.

8. 90-SECOND KILLER DEMO

Time	Action	Judge sees
0–15s	"This is not a medical chatbot. It is a handover firewall."	Product thesis
15–30s	Say: "Patient received 10 mg morphine."	10 mg = HEARD
30–45s	AI starts relay; interrupt: "Wait — it was 5 mg."	TTS stops; correction appears
45–60s	System asks which value is authoritative.	10 mg = SUPERSEDED; 5 mg = CORRECTED
60–70s	Confirm 5 mg.	5 mg = VERIFIED
70–82s	Press Relay / receiving clinician listens.	Only 5 mg is spoken
82–90s	Show state history.	The moat

9. TEAM SPLIT

Person 1 — Voice: STT + Rime + interruption/audio.

Person 2 — Intelligence: LLM extraction + Truth State + conflict logic.

Person 3 — Frontend: dashboard + state visualization.

Person 4 — Integration/Demo: API glue, scenario, fallback flow, pitch.

If time gets tight: prioritize Truth State → conflict engine → voice → dashboard polish.

10. CUT THESE FEATURES

■ Telephony ■ EHR integration ■ Authentication ■ Real patient data ■ Diagnosis ■ Treatment recommendations ■ Fine-tuning ■ Custom STT ■ Complex analytics ■ Huge evaluation suite

KEEP: voice loop + Truth State + contradiction + stale kill switch + verified relay.

11. FINAL PITCH

"Critical handovers don't fail only because information is missing. They fail because information changes.

RELAY-X is a real-time voice handover firewall. It tracks every critical fact as heard, corrected, verified or superseded. When a clinician corrects the AI while it is speaking, we kill the stale response, reconcile the state, and make sure only the verified version reaches the next team.

We don't make AI understand the handover. We make sure the next team receives the right version of it."

Tonight's success condition: One old value gets corrected live, the stale audio stops, the old value becomes blocked, the new value becomes verified, and the receiving team hears only the corrected value. If that works reliably, STOP adding features.